

Rabin-Karp algorithm

```
Rabin-Karp-Matcher(T,P,d,q)
  n = T.length
  m = P.length
  h =  $d^{m-1} \bmod q$ 
  p = 0
  t = 0
  #preprocessing
  for i = 1 to m
    p = (dp + P[i]) mod q
    t = (dt + T[i]) mod q
  #matching
  for s = 1 to n-m+1
    if p == t:
      if  $P[1 \dots m] == T[s, \dots, s + m - 1]$ 
        print "Pattern occurs with shift" s
    if s < n-m+1:
      t = (d(t - T[s]h)+T[s+m])mod q
```

EXAMPLE

```
n : 21
m : 4
d : 10
Divisor, Q : 997
Pattern, P: 2284
p = 0
i = 1 p = (d × p + P[i]) mod Q = (10 × 0 + 2) mod 997 = 2 mod 997 = 2
i = 2, p = (10 × 2 + 2) mod 997 = 22
i = 3, p = (10 × 22 + 8) mod 997 = 228
i = 4, p = (10 × 228 + 4) mod 997 = 290
Text, T : 3124182284769678922843256...
initialize t = 0
h :  $10^3 \bmod 997 = 3$ 
T : 3 1 2 4 1 8 2 2 8 4 7 6 9 6 7 8 9 3 2 5 6
# For the next four steps we are using Horner's rule
# (first for loop in the algorithm),  $t = (d \times t + T[s]) \bmod Q$ 
s = 1,  $t = (d \times t + T[s]) \bmod Q = (10 \times 0 + 3) \bmod 997 = 3 \bmod 997 = 3$ 
s = 2,  $t = (10 \times 3 + 1) \bmod 997 = 31$ 
s = 3,  $t = (10 \times 31 + 2) \bmod 997 = 312$ 
s = 4,  $t = (10 \times 312 + 4) \bmod 997 = 133$ 
# Now we start Rabin-Karp shuffle using rolling hash to compute
# the next substring from the previous substring.
#  $t = (d(t - T[s]h) + T[s+m]) \bmod Q$ 
#  $h = 10^3 \bmod 997 = 3$ 
s = 5,  $(10 \times (133 - 3 \times 3) + 1) \bmod 997 = 244$ 
s = 6,  $(10 \times (244 - 1 \times 3) + 8) \bmod 997 = 424$ 
s = 7,  $(10 \times (424 - 2 \times 3) + 2) \bmod 997 = 194$ 
s = 8,  $(10 \times (194 - 4 \times 3) + 2) \bmod 997 = 825$ 
s = 9,  $(10 \times (825 - 1 \times 3) + 8) \bmod 997 = 252$ 
s = 10,  $(10 \times (252 - 8 \times 3) + 4) \bmod 997 = 290$ 
we found our match
```